

Tutorial 41

M2 returned — and how to read an event study

Advanced Corporate Finance (EBM232A05)

University of Groningen

- **Assessment moment M2:** your reproduction checkpoints come back — 25 points, and more importantly, three weeks of runway to fix what missed.
- Then **event-study craft:** yesterday's lecture leaned on announcement returns (Moeller et al., 2005); today you learn to read such a table the way you read a regression table — as **target cells with choices behind them.**

M3 — the final package — is due at the end of **next** week. The reconciliation work starts from today's feedback.

M2 — what came back

Reading your M2 feedback

- **D1a (20, scripted)**: per target cell — full ($\pm 2\%$ + sign and significance), partial ($\pm 20\%$), low ($\pm 40\%$). The harness diff report shows each cell.
- **D1b (5)**: funnel internal consistency and whether each $> \pm 2\%$ miss is **root-caused** — data vintage / underspecification / your choice / suspected paper error, with evidence.
- A miss now is **not** a lost cause: cells re-scored at M3 stand on the final package. What is fatal is a miss you cannot *explain*.

The recurring pattern: coefficient in band, **SE off by half** — almost always clustering. Second place: *N* off by thousands — almost always a sample filter the paper states in a footnote. Check those two before anything else.

Event-study craft

The design, in one frame

The question: what did the **announcement** do to the price? The answer requires a counterfactual — what the stock *would* have returned anyway:

$$AR_{it} = R_{it} - \underbrace{(\hat{\alpha}_i + \hat{\beta}_i R_{mt})}_{\text{market model, fit in the estimation window}}, \quad CAR_i[-1, +1] = \sum_{t=-1}^{+1} AR_{it}.$$

- **Estimation window** (e.g. -205 to -6): fit the counterfactual. **Event window** (-1 to $+1$): measure the surprise. They must not overlap.
- The design works because news arrival is **dated**: in an efficient market, a short window isolates the announcement from everything slow-moving.

Reading the table = the same craft as week 38

An event-study table's cells are target cells like any regression's — with the same silent choices behind them:

- **Which window?** $[-1, +1]$ vs longer — longer admits confounding events.
- **Which benchmark?** Market model vs market-adjusted vs factors — tenths of a percent; matters when the effect is small.
- **Which mean?** Equal-weighted says “the average deal”; **dollar-weighted** says “the average dollar” — Moeller et al. (2005)'s whole point: small deals gain, huge deals destroy ($-\$397\text{bn}$ in 87 deals; $\$2.31$ per dollar spent), so the average deal is fine while aggregate dollars are not.

What a CAR can and cannot tell you

- It measures the market's **revision of beliefs** about the acquirer — deal value *and* what the bid reveals about management, standalone value, and overpayment, bundled.
- Yesterday's lecture: acquirer CARs ≈ 0 fits **both** the competition story and the free-rider story — the observed deals are the **selected** ones that cleared the appropriability hurdle. An announcement return cannot settle what the population of *possible* deals looks like.
- For your paper: if your target table is an event study, your funnel has an extra stage — **event selection** (which announcements enter) — and it is where reproductions usually diverge.

Before you leave

- **This week:** work through your M2 feedback; every missed cell gets a root cause in the final reconciliation.
- **Next week:** Tuesday, the risk-management lecture. Wednesday, **Problem set 42 (P1–P4)** and the **M3 checklist** — the final package is due at the end of week 42.
- **Week 43:** individual reflections due at the start of the week; the **code defence** in the tutorial. Every member, every part of the repo.

Moeller, S. B., F. P. Schlingemann, and R. M. Stulz (2005). **“Wealth Destruction on a Massive Scale? A Study of Acquiring-Firm Returns in the Recent Merger Wave”**. In: *The Journal of Finance* 60.2, pp. 757–782.